

A Node with a Granularity of NoTimeGranularity SHALL attempt to sync its time at least once a day. Nodes SHOULD NOT query the Time Synchronization cluster of another Node more than once per 30 minutes.

11.17.15. Acting as an NTP Server

Any capable Node with always-on power source that has and can maintain a time synchronization Granularity of MillisecondsGranularity or better, SHOULD act as an NTP server. Any capable Node that reaches the final stage in the [Time source prioritization](#) mechanism SHOULD act as an NTP server for the network.

A Node hosting an NTP server SHALL update the NTPServerAvailable attribute and SHOULD advertise an _ntp._udp DNS-SD service. A Node hosting an NTP server SHOULD support network time security (NTS). A Node hosting an NTP server SHOULD implement leap smearing.

11.17.16. Implementation Guidance

This specification offers several options for getting time in order to support Nodes with various capabilities. This section is not intended as a requirement, but is used to illustrate how various Nodes might implement this specification.

11.17.16.1. Example: Constrained Node with no schedule capabilities

This type of Node would not need to support the TimeZone (TZ) feature or the NTPServer (NTPS) feature. A constrained node can choose to get time either by using a time synchronization client cluster to read the time from a trusted node (if the TSC feature is supported), and / or by using the NTPClient (if the NTPC feature is supported). A constrained node supporting NTPC would likely use an SNTP implementation.

A node supporting both the NTPC and TSC features would have the following attributes:

- UTCTime
- Granularity
- TimeSource (optional)
- TrustedTimeSource
- DefaultNTP

To achieve time synchronization, the Node would start at Step 3 of [Time source prioritization](#) (Steps 1 and 2 do not apply).

If the Node opts to support only the TSC feature, it would have the following attributes:

- UTCTime
- Granularity
- TimeSource (optional)
- TrustedTimeSource